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Photochemically generated ionic species for cationic and step-growth polymerizations

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ABSTRACT

Developing innovative photochemical methods for the purpose of attaining green chemistry and mitigating global warming has recently become a very important concept. While we have to move towards a more sustainable world, daily life is becoming increasingly dependent on the use of polymeric materials. Conventional thermal polymerization systems fail to fulfill the requirements of sustainable chemistry, resulting in an increase in energy demand. This article outlines cationic and step-growth polymerizations of various monomers initiated by different photochemically generated ionic species and their advantages over conventional polymerization systems. Typical examples of photoinduced cationic and step-growth polymerizations performed under UV, visible and near-infrared (NIR) light are summarized under two main categories; direct and indirect initiating systems. In direct initiating systems the use of commercially available and structurally modified onium salts as one-component photoinitiators for both polymerizations are summarized. Indirect initiating systems, proceeding through i) photoinduced electron/energy transfer (PET), ii) charge-transfer complexation and iii) free-radical promoted cationic polymerization (FRPCP), are summarized with a focus on the recent step-growth methods developed in our laboratory.

1. Introduction

Photoinitiated polymerization reactions have received considerable interest, especially over the last two decades, due to several advantages over conventional techniques such as low energy consumption, high rates of polymerization and the use of less amount of solvent [1–4]. Photopolymerization reactions occur only in the illuminated areas providing spatiotemporal control which is not possible to achieve by conventional polymerization methods. The control of space and time is crucial for many high-technology applications like 3D-printing, fabrication of conductive films, microcircuits and dental fillings [5–9]. Recently, photoinduced methods have been increasingly applied in controlled/living polymerizations mostly in atom transfer radical polymerizations (ATRP), radical addition-fragmentation chain transfer polymerizations (RAFT) and "click reactions" producing well-defined architectures and functionalities [10–18].

Photopolymerization is basically a solidification process that transforms monomers into polymers through reactions initiated by radicals or ions, which are generated from photosensitive compounds (PIs) upon irradiation in a broad range of electromagnetic spectrum depending on the spectral sensitivity of the initiating system. Visible and NIR activations are highly preferred due to their environmentally friendly and less harmful nature. Particularly, NIR light has several advantages compared to visible light including lower energy requirement, deeper penetration capability and additional heat release which favors the polymerization processes [19–21].

Although a great majority of the photopolymerization reactions deal with free-radical polymerization (FRP), cationic photopolymerization offers many advantages over FRP including no oxygen inhibition, dark polymerization, low volume shrinkage, better clarity, adhesion and chemical resistance [22–24]. Special active centers which are formed during cationic photopolymerizations are theoretically non-terminating and have extremely long active center lifetimes. This phenomenon is called "dark polymerization" [25]. Besides the cationic photopolymerization of classical monomers such as vinyl ethers and oxirane derivatives, lately, focus has been placed on the photoinduced step-growth polymerization of aromatic monomers, mainly thiophene, pyrrole, pyrene, carbazole, indole and their derivatives due to their wide range of applications for the fabrication of many high-tech devices [26–32]. These polymers can be formed through the reactions of their

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[1] This paper is dedicated to the memory of our dear professor Yusuf Yagci, who passed away while this paper was being peer-reviewed.

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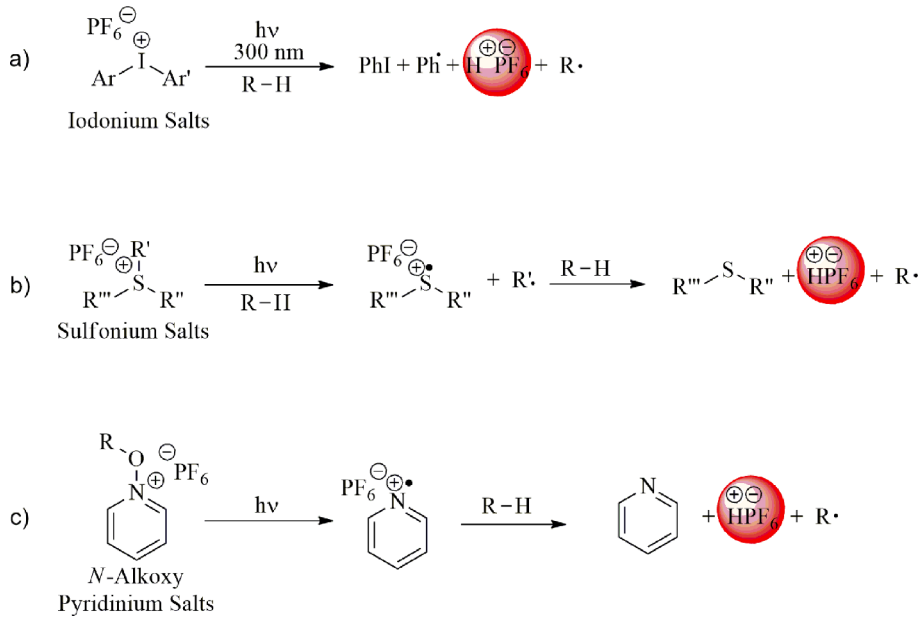
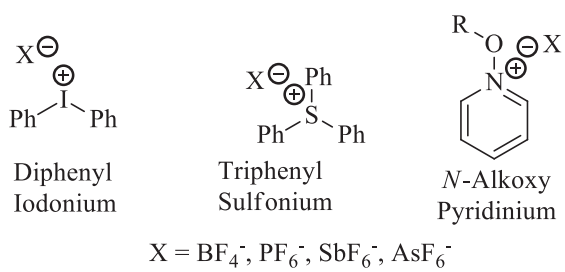
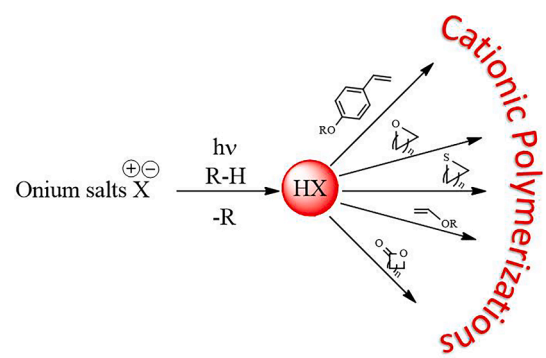


Scheme 1. Photogeneration of Brønsted acids using onium salts.



Scheme 2. Commonly used onium salts for the photocationic polymerization reactions.



Scheme 3. Photoinduced cationic polymerizations of cycloethers, cyclothioethers, epoxides, vinyl ethers, lactones and styrenes induced by onium salts [34].

photochemically generated ionic monomers (radical cations) capable of initiating step-growth polymerizations by subsequent reactions. The aim of this article is to summarize the recent approaches for photoinitiated cationic and step-growth polymerizations based on different direct and indirect initiating systems with a special emphasis on the photopolymerization methods of aromatic monomers recently reported by our group.

2. Direct initiating systems

In direct initiating systems for cationic polymerization, the light energy is directly absorbed by photoinitiators (PIs) mainly by onium salts [33–35]. These light sensitive ionic compounds are formed from the elements of groups 5A, 6A, 7A with a bulky, inorganic counter anion. The photodecomposition of onium salts with high quantum yields can generate Lewis acids capable of initiating the cationic polymerization of various monomers including cyclic monomers and vinyl ethers [36].This is the reason why most of the onium salts are also called photoacid generators (PAGs)[37,38] (Scheme 1).

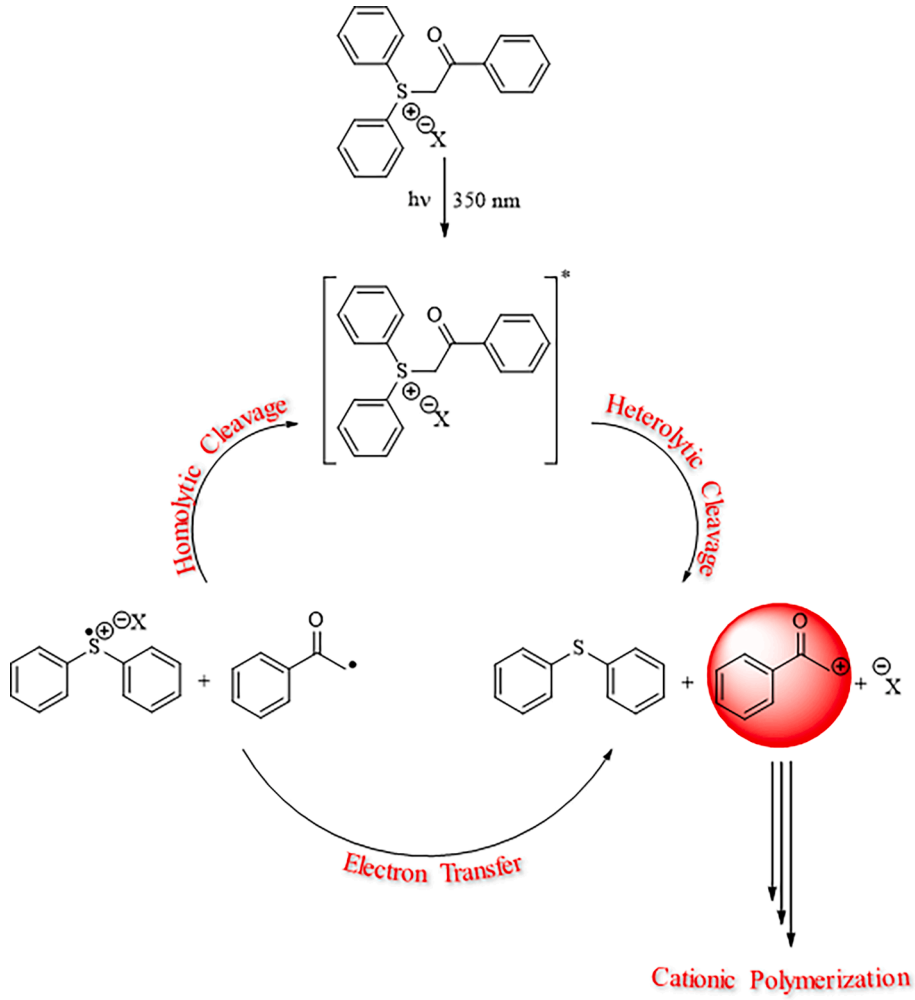
Depending on their spectral sensitivity, PAGs can be used under different wavelength irradiation for the generation of ionic species to initiate cationic or step-growth polymerizations of cyclic monomers [39]. Most of the PAGs exhibit good thermal stability, high solubility in polar monomers and high photoinitiation efficiency [40,41]. In regard to the mechanism of the cationic polymerization induced by PAGS, cation is the light-absorbing species associated with the photochemistry process. It has a crucial role in the thermal stability, UV absorption and photoefficiency of the PAG. The counter anion determines the acid strength, hence, the initiation efficiency of polymerization reaction. Thus, the polymerization rate is directly related to both nucleophilicity and size of the counter anion. Low nucleophilicity of the counteranion favors the generation of strong acids. According to previous studies, molecular weights and percentage conversion of cationically polymerizable monomers increase in the order of $BF_4^- < PF_6^- < AsF_6^- < SbF_6^-$ [34]. Scheme 2 shows some of the commonly used onium salts for the photocationic polymerization reactions.

2.1. Direct cationic photopolymerization using onium salts under UV light

The first cationic photopolymerization reactions were reported by Crivello and Lam in 1977 [34]. The reactions were initiated by the irradiation of iodonium salts under UV light (Scheme 3).

Structurally different onium salts, including iodonium, sulfonium, pyridinium, phosphonium, anilinium and pyrylium have extensively been studied and applied in both academic and industrial domains for the polymerization of various types of monomers [42–47].

A major drawback of many onium salts is their low spectral sensitivity which is most of the time limited to UV region. Phenacylium salts are a class of onium salts that overcomes the low spectral sensitivity problem to some extent by the incorporation of a chromophore group



Scheme 4. Photoinduced homolytic and heterolytic cleavages of diphenyl phenacyl sulfonium salt initiating cationic polymerization [48].

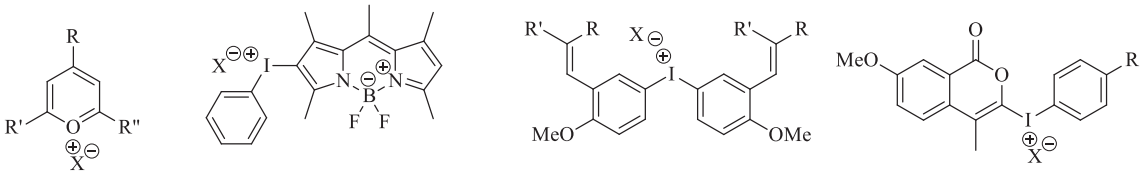
(phenyl ketone group) which slightly shifts the absorption to longer wavelengths [48]. Phenacylium salts can be facilely synthesized by reacting phenacyl halides with the corresponding heteroatom nucleophiles such as dialkyl(aryl)sulfide followed by the subsequent ion exchange with the sodium, potassium, or silver salt of a non-nucleophilic counter anion. Two main photocleavage mechanisms were proposed (Scheme 4).

While homolytic cleavage forms a phenacyl radical and the corresponding heteroatom radical cation, heterolytic cleavage results in the formation of the phenacyl cation. In the former case, internal electron transfer also yields phenacylium cation. Thus formed ionic species by two different pathways were shown to initiate cationic polymerization under near-UV light (UV-A) which has slightly lower energy than UV-B and UV-C light which are used for most of the classical onium salts.

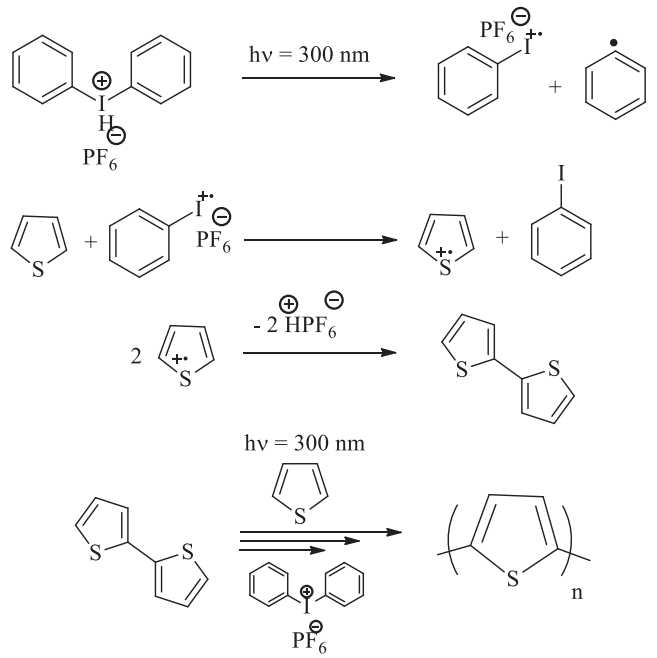
2.2. *Direct cationic photopolymerization using onium salts under visible light*

As briefly explained in the previous section, a major drawback associated with onium salts is their low spectral sensitivity which limits activation beyond high energy UV region (UV-C). Most of the phenacylium salts don't have absorbance in the visible region. Recently, there has been an immense interest in the use of onium salts for direct cationic polymerization under visible light irradiation. Recently, Ortyl, Lalevée and other groups have been able to design, synthesize and investigate the efficiency of pyrylium salt and new iodonium salts containing different chromophore groups such as coumarin, BODIPY and benzylidene as one-component cationic photoinitiators absorbing in the visible region up to 600 nm (Scheme 5) [49–55].

The synthesized iodonium salts were highly active in the photopolymerization of epoxides and vinyl ethers under visible light light-emitting-diode (LED) irradiation. The detailed investigation of photopolymerization kinetics through electron spin resonance, laser flash



Scheme 5. One-component cationic photoinitiators reported by several research groups.



Scheme 6. Photoinduced step-growth polymerization of thiophene using DPI [30].

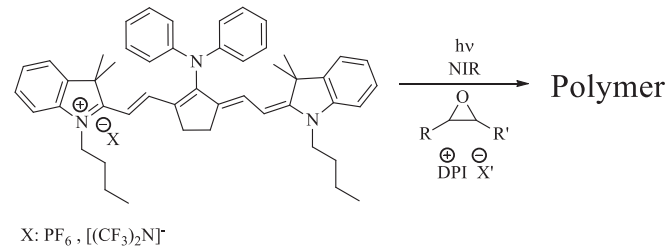
photolysis experiments and the cyclic voltammetry studies showed that the highest efficiencies were reached using anions with highest nucleophilicities and in the presence of donor-π- acceptor nature in the chromophores. Hence, the cation part was very important for the photoinitiator's high absorption and high catalytic activity.

2.3. Direct Step-Growth polymerization using onium salts

It was first demonstrated by our group in 2005, that polythiophene can be obtained under UV irradiation (~300 nm) using an onium salt, namely, diphenyl iodonium hexafluorophosphate (DPI) as one-component PI [56]. Later in 2007, the detailed reaction mechanism was proposed by taking advantage of laser flash photolysis and electron paramagnetic resonance (EPR) studies [30]. The mechanism of the photoinduced step-growth polymerization of thiophene is similar to the already-known chemical or electrooxidative polymerization mechanism. Photolysis of iodonium salt forms initially PhI-$^{+}$ which undergoes electron transfer resulting in the formation of thiophene radical cation. Subsequent proton release and coupling reactions form thiophene dimer. Successive three processes essentially yield polythiophene. The overall mechanism is presented in Scheme 6.

3. Indirect initiating systems

Regarding indirect initiating systems, there are three main modes for the photochemical generation of ionic species for the initiation of cationic or step-growth polymerizations: i) photoinduced electron/energy transfer (PET) ii) charge-transfer complexation iii) free-radical promoted cationic photopolymerization (FRPCP).



Scheme 8. NIR light induced cationic polymerization by cyanine dyes and DPI.

3.1. Cationic photopolymerization by PET under visible light

Besides structural modification method, the spectral sensitivity of onium salts can also be extended to longer wavelengths (visible/NIR region) by the use of different light sensitive compounds, namely, photosensitizers (PS), which do not directly initiate the polymerization reactions but plays a crucial role in the photoinduced energy/electron transfer reactions [57–62].

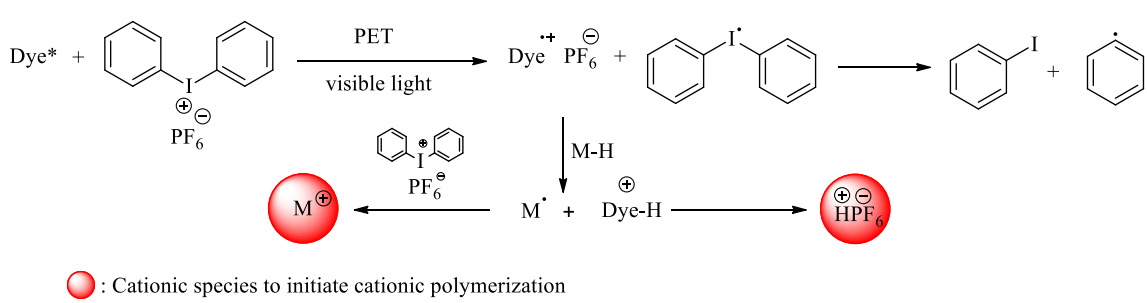
In this approach the spectral sensitivity of PS governs the overall absorption character of the photoinitiating system. Dyes having different structures and a broad range of light sensitivity has been efficiently used as PS in conjunction with diphenyl iodonium hexafluorophosphate (Scheme 7) [63].

3.2. Cationic photopolymerization by PET under NIR

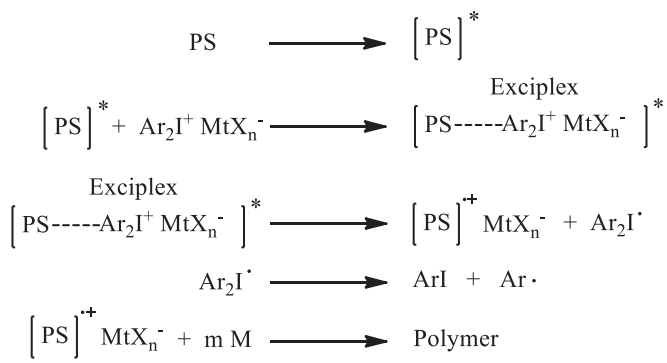
Many of the modern industrial applications including 3D printing and digital imaging (DI) involve the use of cheap and safe NIR light due to its deeper penetration ability into the matter. In addition to this, the heat release by NIR irradiation can enhance the efficiency of the polymerization processes, a phenomenon called as "photothermal effect". For this purpose, different PSs that can absorb NIR light have been used in conjunction with onium salts [64,65]. Organic dyes, especially cyanines, are typical NIR photosensitizers for this purpose. Pioneering work by Strehmel group demonstrates the possibility of NIR-sensitized cationic polymerization by using cyanine dyes. The efficiency of several cyanine dyes with different substituents (electron donor or electron acceptor) as PSs for diphenyl iodonium salt in the cationic polymerization reaction of oxiranes using high-power LEDs were investigated (Scheme 8) [66].

3.2.1. Step-Growth polymerization by PET

Especially, typical polynuclear aromatic hydrocarbons (PAH) such as pyrene are widely used as PS for the near-UV photodecomposition of onium salts via electron transfer in the excited complex, coined as exciplex [67]. In this complex, one electron is transferred from the



Scheme 7. Visible light induced generation of cationic species by using dyes.



Scheme 9. Photoinduced cationic polymerization using photosensitizers.

sensitizer molecule to the onium salt, giving rise to the generation of sensitizer radical cations and neutral onium radical species Thus formed radical cations can initiate cationic polymerization of typical cyclic monomers or vinyl ethers. These radical cations can also abstract hydrogen from donor molecules to generate Brønsted acids that are also capable of initiating cationic polymerizations (Scheme 9).

It is important to emphasize here that in the absence of a vinyl monomer, the radical cations of PS, generated by PET between PS and onium salt, especially when the concentration of PS is high, can self-couple through release of HX acid forming dimer as proposed by Crivello (Scheme 10) [67].

Under certain conditions readily formed dimer then can couple with another dimer or monomer forming higher molecular weight species in a step-growth manner. Table 1 shows different cationic photoinitiators with corresponding cation generation routes and wavelength characteristics.

3.2.2. Step-Growth polymerization by PET under UV and visible light

The photoinduced step-growth polymerization method used for thiophene was further extended to obtain other technologically important polymers such as poly (*N*-ethylcarbazole), polythienothiophene derivatives, polypyrene, poly (*N*-methylpyrrole), poly (*N*-methylindole) and poly(ethylenedioxy thiophene) (PEDOT) using DPI and phenacyl bromide as one-component photoinitiator under near-UV and visible light irradiation (Scheme 11) [26,32,68–72].

PET reaction of the excited thiophene derivatives or *N*-ethylcarbazole (EC) in the presence of oxidizing agent proceeded efficiently giving monomer radical cations ($M^{+\bullet}$). Subsequent two protons release and coupling of two monomer radical cations generate dimers. Consecutive similar reactions concerning excitation, electron transfer, proton release, and coupling finally lead to the formation of the respective conjugated polymer.

3.2.3. Step-Growth polymerization by PET under NIR

The above explained photoinduced step-growth polymerization by PET methods also succeeded under NIR for the photopolymerization of a donor–acceptor-donor (DAD) type monomer, thiophene substituted [1,2,5] chalcogenazolo[3,4–*f*]-benzo [1-3] triazole (chalcogen = S, Se) [73].

Since the monomers absorb the light at NIR region where DPI is transparent, the proposed mechanism involves the same sequence of the reactions described for UV–visible systems as depicted in Scheme 12. Due to the donor–acceptor-donor nature, the polymers resulted in very low optic and electronic band gaps which showed the efficiency of the developed method for the synthesis of DAD-type polymers.

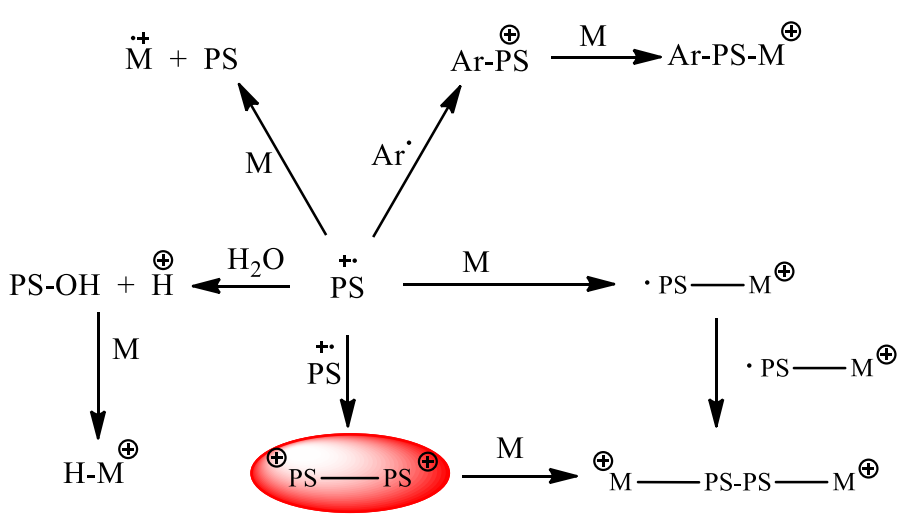
3.3. Cationic photopolymerization by Charge-Transfer complexation

Ground state charge-transfer complexes (CTCs) [74] formed between electron donor and acceptor (EDA) molecules appear to be a versatile photoinitiating system for cationic polymerization [75]. Formation of CTCs depends on the electronic interactions between the lowest unoccupied molecular orbital (LUMO) of the acceptor molecule and the highest occupied molecular orbital (HOMO) of the donor molecule where the energy gap of HOMO-LUMO is smaller than that of initial donor and acceptor molecules independently. This phenomenon, most of the time, causes a significant bathochromic shift (red-shift) in the light absorption compared to that of parent molecules [76]. Upon irradiation, radical cations are formed by electron transfer the excited CTC. The followed initiating protonic acid generation is similar to the previous cases (Scheme 13).

The main benefits of using CTCs are i) the ability to extend the excitation wavelength, thus, performing the photopolymerization under lower energy conditions; ii) the possibility of dual (photo/thermal) polymerization since most of the CTCs can also form active species (radicals/ions) when heated; iii) low optical density maximizing penetration ability of light that favors formation of thick samples through excellent curing depths [74].

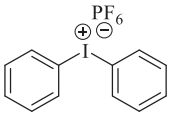
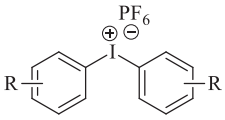
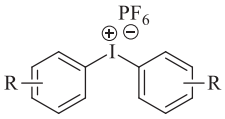
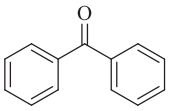
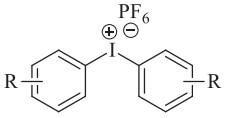
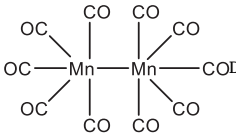
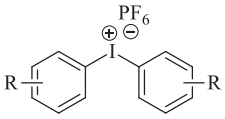
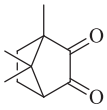
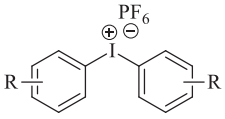
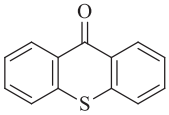
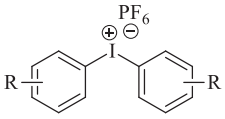
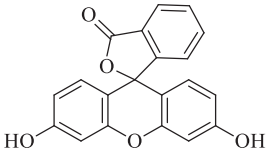
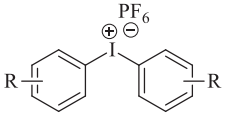
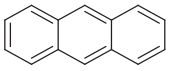
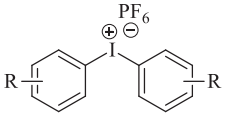
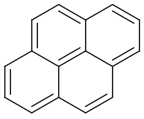
In CTCs, the nucleophilicity of the counter anion has a great impact in photocationic polymerization as they are responsible for the formation of cationic species. Solvent effect is another important parameter for the strength of CTCs. Typically, as the polarity of the solvent is proportionally associated to the strength of CTC [77].

CTCs between alkoxy pyridinium salts and aromatic electron donors



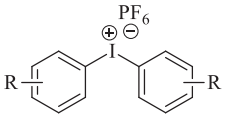
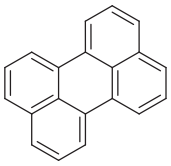
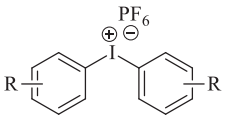
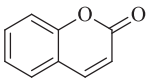
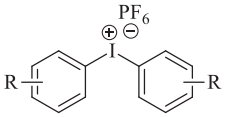
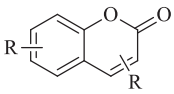
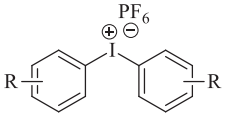
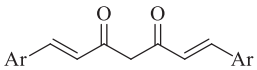
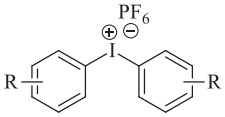
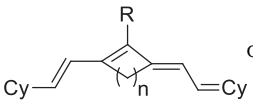
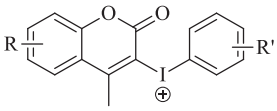
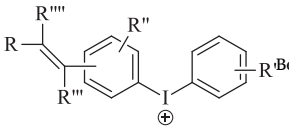
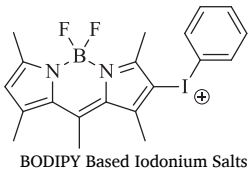
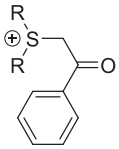
Scheme 10. Different pathways for photoinduced cationic polymerization involving photosensitizer radical cation [67].

Table 1
Several photoinitiating systems (PIS), their corresponding cation generation routes and excitation wavelengths.

Photoinitiator	Photosensitizer	Cation Generation Route		Excitation Wavelength		
		Direct	Indirect	Absorption Range (nm)	λ max (nm)	Spectral Region
	–	✓		Up to 305	250	UV-B
 R: Heteroatomic or aromatic groups	–	✓		Up to 310	275	UV-B
 R: Aliphatic and heteroatomic groups	 Benzophenone		✓	Up to 355	255–330	UV-A
 R: Aliphatic and heteroatomic groups	 Dimanganesedecacarbonyl		✓	Up to 450	350–390	Visible
 R: Aliphatic and heteroatomic groups	 Camphorquinone		✓	350 to 525	455	Visible
 R: Aliphatic and heteroatomic groups	 Thioxanthone		✓	Up to 430	260–390	Visible
 R: Aliphatic and heteroatomic groups	 Fluorescein		✓	Up to 550	510	Visible
 R: Aliphatic and heteroatomic groups	 Anthracene		✓	Up to 360	320–340	UV-A
 R: Aliphatic and heteroatomic groups	 Pyrene		✓	Up to 360	320–340	UV-A

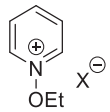
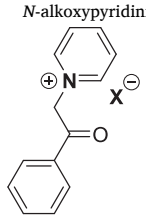
(*continued on next page*)

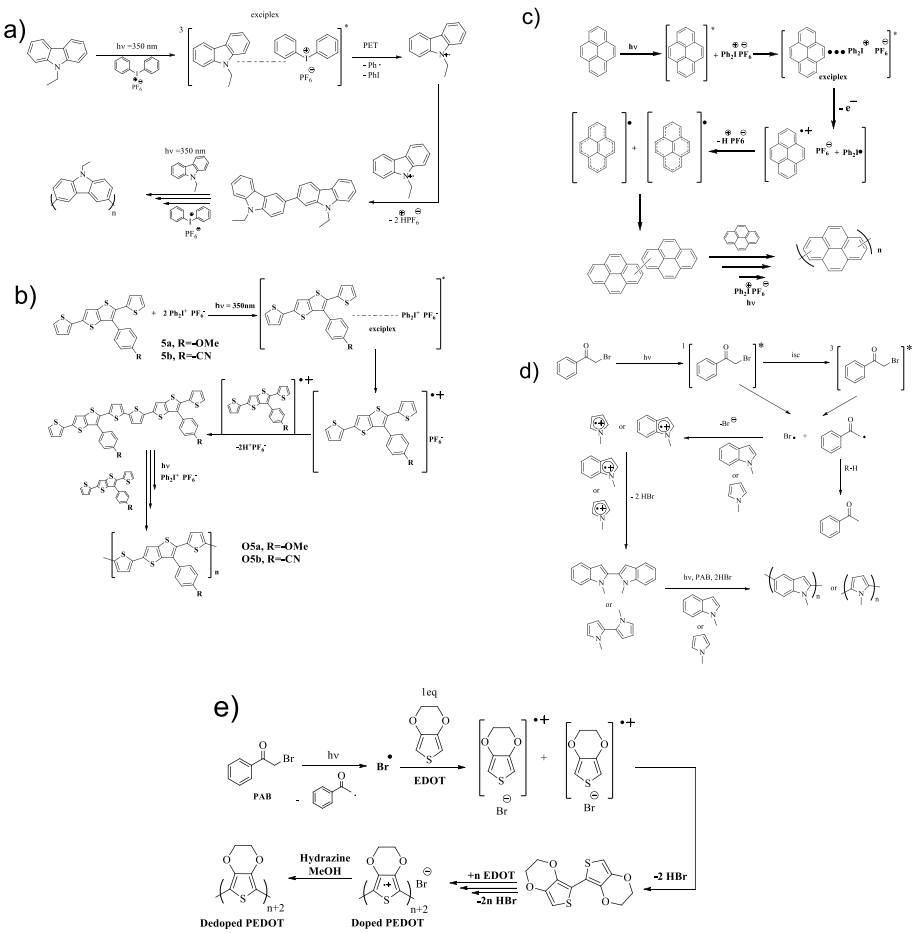
Table 1 (*continued*)

Photoinitiator	Photosensitizer	Cation Generation Route		Excitation Wavelength		
		Direct	Indirect	Absorption Range (nm)	λ max (nm)	Spectral Region
 R: Aliphatic and heteroatomic groups	 Perylene		✓	350 to 460	410–435	Visible
 R: Aliphatic and heteroatomic groups	 Coumarin		✓	Up to 345	275–320	UV-A
 R: Aliphatic and heteroatomic groups	 Substituted Coumarins		✓	Up to 525	342 to 445	Visible
 R: Aliphatic and heteroatomic groups	 Curcuminoids		✓	Up to 540	384 to 454	Visible
 R: Aliphatic and heteroatomic groups	 Cyanine Dyes		✓	600 to 850	800	Visible to NIR
 Coumaryliodonium Salts	–	✓		Up to 400	375	Visible
 Based Iodonium Salts	–	✓		Up to 450	Up to 400	Visible
 BODIPY Based Iodonium Salts	–	✓		Up to 600	Up to 500	Visible
Triarylsulfonium Salts	–	✓		Up to 310	280	UV-B
 Phenacylsulfonium Salts	–	✓		Up to 355	Up to 280	UV-A

(*continued on next page*)

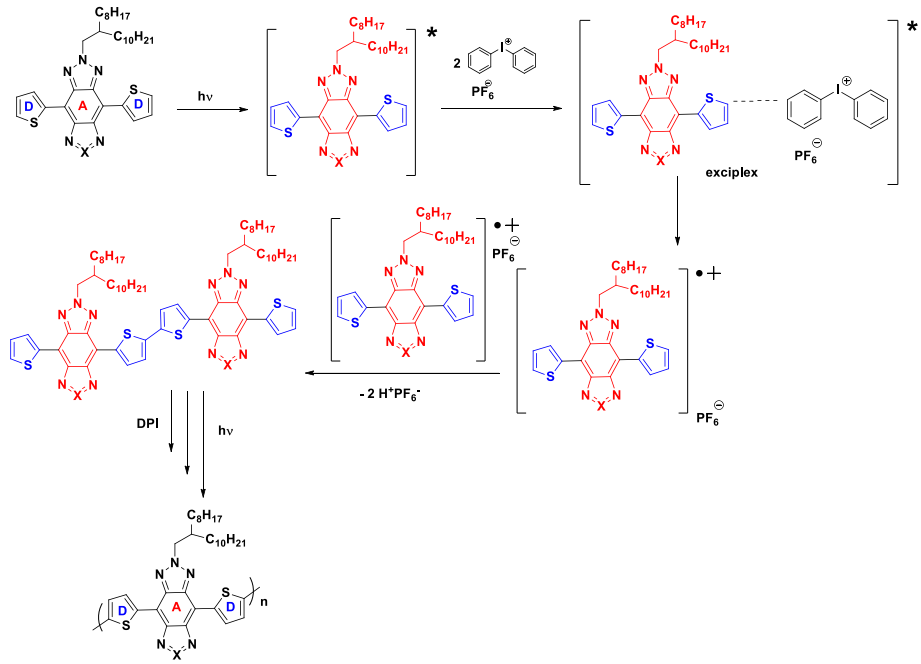
Table 1 (*continued*)

Photoinitiator	Photosensitizer	Cation Generation Route		Excitation Wavelength		
		Direct	Indirect	Absorption Range (nm)	λ max (nm)	Spectral Region
 *N*-alkoxypyridinium Salts	–	✓		Up to 355	Up to 305	UV-A
 *N*-phenacylpyridinium Salts	–	✓		Up to 483	Up to 455	Visible to UV-A

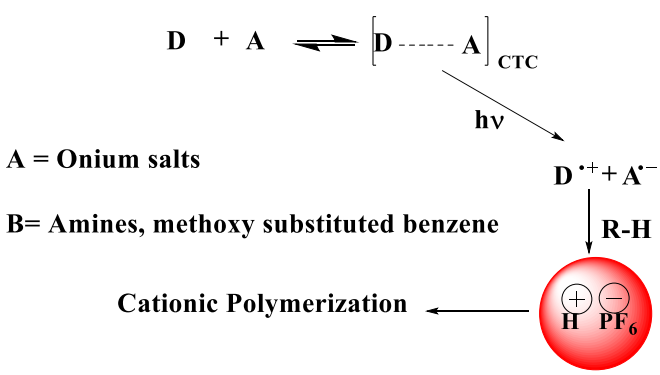


Scheme 11. Photoinduced step-growth polymerization of (a) *N*-ethylcarbazole, (b) thienothiophene derivatives, (c) pyrene and (d) *N*-methylpyrrole, *N*-methyindole e) EDOT.

such as methoxy substituted benzene were previously shown by our group to initiate cationic polymerization under near-UV and visible light [78]. Recently, visible light CTC initiators based on the interactions between sulfonium or iodonium salts as electron acceptors and amines or phosphines as electron donors were reported by Lalevée and our group [79–81] Sulfonium salts generally have higher thermal stability and are less hazardous than iodonium salts. Dual initiating ability of a sulfonium salt-based CTC was also demonstrated [81]. In addition, sulfonium salts are preferred over iodonium salts due to the better solubility in most of the cationically polymerizable monomers such as vinyl ethers and heterocyclic monomers (Scheme 14) [82].



Scheme 12. Photoinduced step-growth polymerization of DAD-type monomers under NIR light.



Scheme 13. Photoinitiated cationic polymerization by charge-transfer complexation.

4. Free-radical promoted cationic photopolymerization and its applications in Step-Growth polymerization

Free-radical promoted cationic polymerization (FRPCP) was first reported as early as in 1978 [83]. This innovation opened a new horizon for the development of several cationic routes as photochemically or thermally generated radicals can readily be oxidized by onium salts to the corresponding cations to initiate cationic polymerizations [84–87]. A vast number of sources emitting light at different wavelengths can be utilized for FRPCP including UV, visible or NIR light.

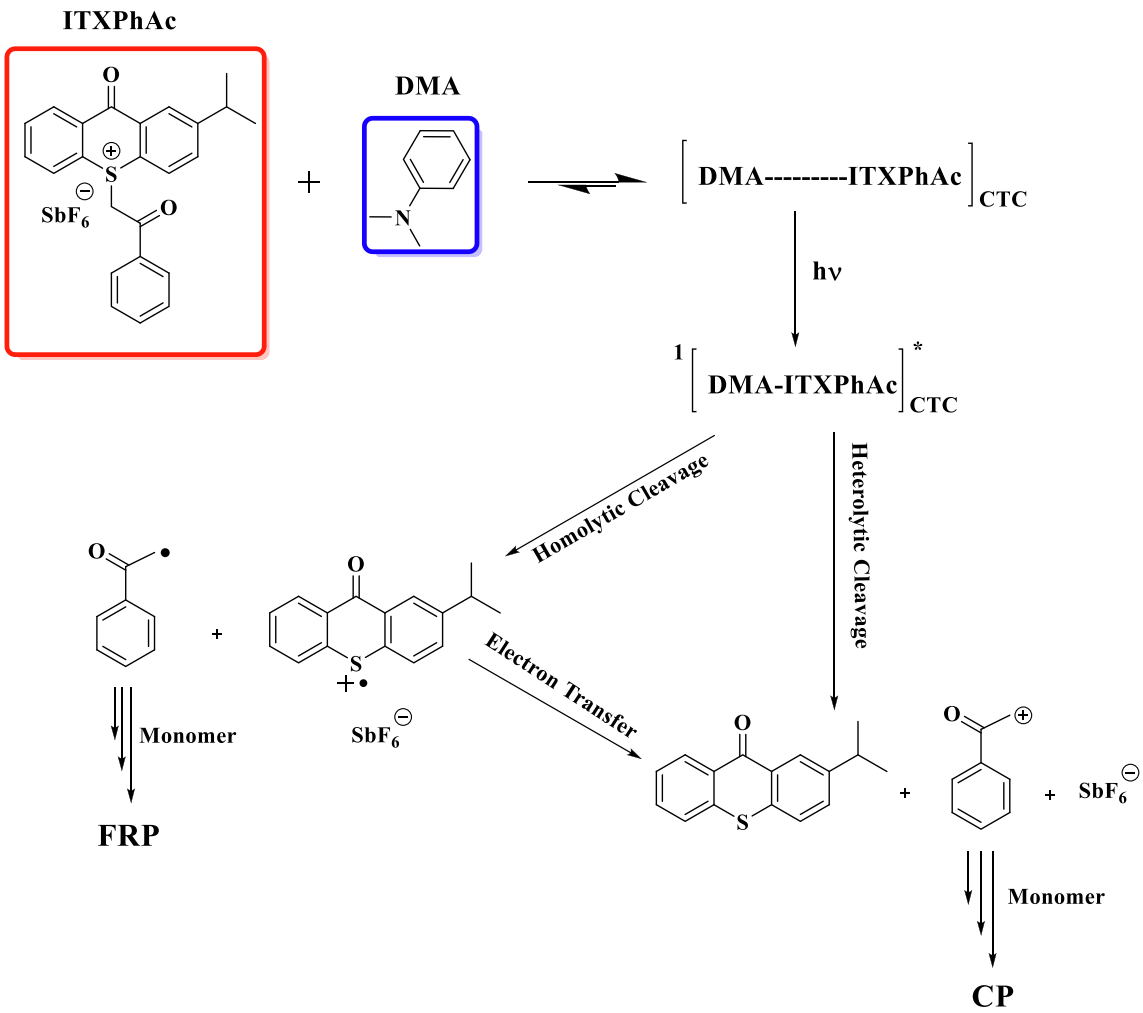
4.1. FRPCP under UV light

Initial studies were conducted by using UVA absorbing PIs. The electron donor free radicals formed from well-known PIs such as benzophenone or acetophenone derivatives. This way, the absorption characteristic of the initiating system is extended to longer wavelengths. Moreover, the polymerizations can also be initiated by thermal initiators (Scheme 15) [88–91].

4.2. FRPCP under visible light

FRPCP was proven to be useful to broaden the spectral response of onium salts by several research groups [92–94] Vast options of photoinitiators and oxidizing salts allow the researchers to tune the absorption wavelength of photopolymerization [95–98] The main commercially available visible light photoinitiators including acyl germanes, acyl phosphine oxides and acyl silanes that undergo homolytical cleavages after irradiation to yield radicals. This type of initiators are coined as *Type I* photoinitiators [99–102]. However, some of the resulting radicals generated by photocleavage are not capable to be oxidized by onium salts. Instead, they may have a tendency to abstract hydrogen or halogen. The resulting radicals are oxidizable to promote cationic polymerization [103–111] (Scheme 16).

Recent advances involve the use of mentioned techniques in the field of additive manufacturing (AM). AM technology is being developed rapidly to yield higher-resolution products and the device's costs are lowering every day and thus, usable for [18,112–120]. These devices mainly involve the use of visible light in such systems to benefit from spatiotemporal control and the relatively safe nature of visible light. As the commercial availability of visible light LEDs is increased, the production costs are directly affected. Photoinduced AM techniques involve both radical and cationic polymerizations. Since the presence of aerial oxygen in the reaction environment has a negative effect on the printout resolution and employing an inert atmosphere is industrially challenging, advanced approaches were developed in order to overcome this issue in photoinduced RAFT, ATRP and FRPCP approaches [18,121–124]. Some of the reported photo RAFT and ATRP techniques yield not only high-resolution printouts and also surface modification and nanoscale printings by Xiao, Boyer, Zhu and Lalevée research groups [125–131]. FRPCP systems are being utilized to these systems by several research groups in order to obtain high-resolution printouts with resins that are only polymerizable by cationic routes [132] These types of resins are proven to be stable under air and possess storage stability for long time (Scheme 18) [133]. Additional to AM, frontal polymerization techniques (FP) may require specialized equipment and irradiation with UVA and UVB light too. Although directly induced cationic FP techniques are investigated thoroughly, in order to extend the excitation wavelength to the visible light region FRPCP approaches also are



Scheme 14. Photoinitiated hybrid polymerization by the charge-transfer complexation between DMA and ITXPhAc [77].

involved in FP [134,135]. The technique mainly stands out as a self-sustained reaction in which, once the polymerization has been initiated from a point, it propagates through the matrix until the end as the process is highly exothermic. In one of the initial efforts to combine FRPCP with FP, Liska group successfully combined the perbenzoic acid type radical initiators and epoxy monomers with DPI salts to obtain FP [134]. Recent approaches by Sangermano involve bio-based epoxy monomers and iodonium salt derivatives as oxidizers in conjunction with tetraphenyl ethanediol as a radical initiator (Scheme 17) [135,136].

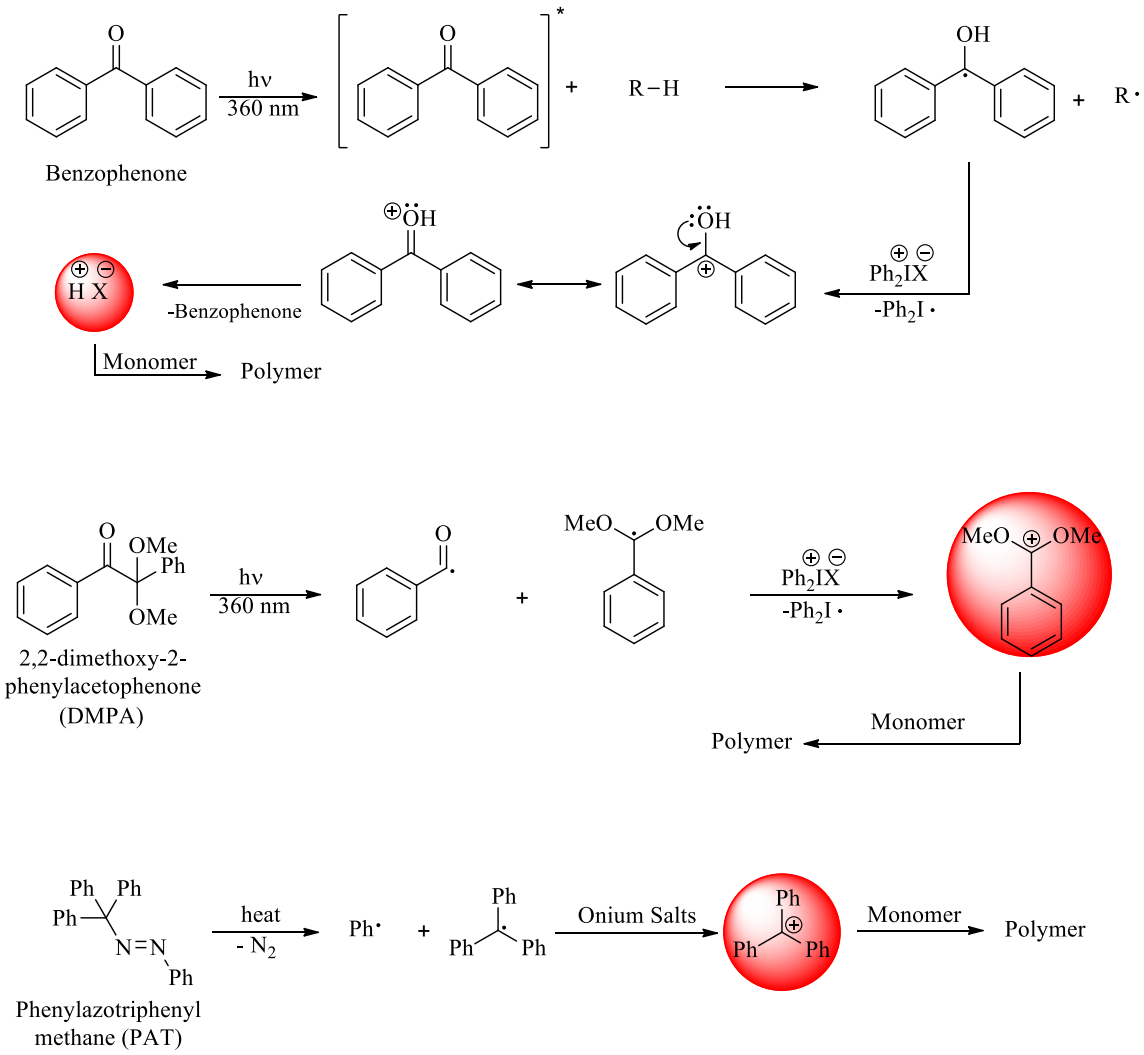
4.3. FRPCP under NIR light

Utilization of NIR irradiation in polymerization systems is advantageous due to the lower energy requirement matching with the global concerns. Additionally, the overall efficiency of the reaction is increased and operational safety is improved. In curing related applications, NIR irradiation provides deeper penetration through materials and the overall efficiency of energy conversion is greater compared to the systems operating with UV and visible light. NIR-induced cationic photopolymerizations can be realized by using several approaches including upconverting nanoparticles (UCNP) and NIR absorbing photosensitizers [19,137–143].

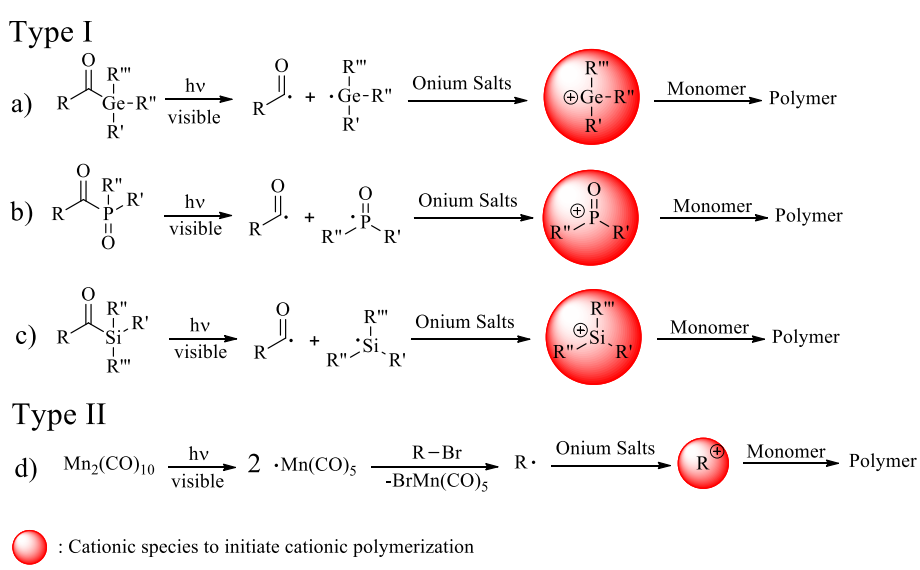
In one of the initial approaches involving UCNP, our research group successfully developed three component FRPCP system involving UCNP, titanocenes, and DPI [137] Commercially available titanocenes were excited by the emission of UCNPs therefore generating radicals to be oxidized by DPI salts initiating the cationic reaction (Scheme 18).

4.4. Photinduced Step-Growth polymerization using FRPCP

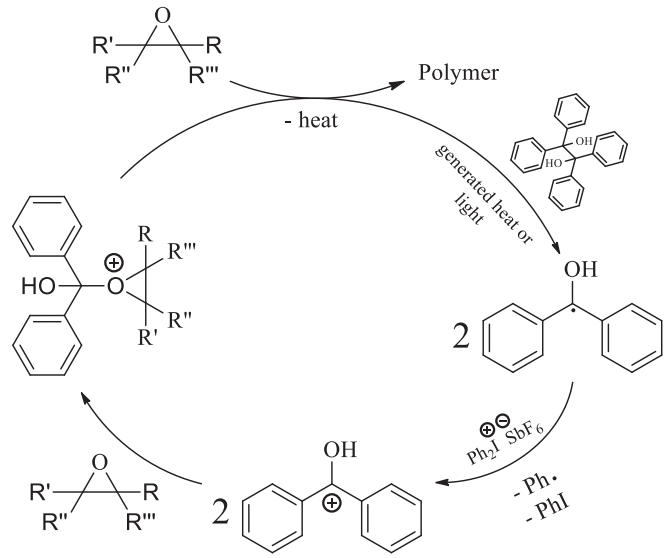
In addition to other advanced approaches, there are also methodologies that involve FRPCP to obtain blocks, grafts and chain extensions [144–146] Prepolymers obtained by several techniques including radical, or cationic routes may be used as a macroinitiator to form blocks and functionalized side chains may yield grafts by FRPCP [147] Also cationic species can be directly utilized in advanced reactions including photoinduced radical oxidation addition deactivation process (PROAD) or Friedel-Crafts type electrophilic aromatic substitution reactions [148,149] As photochemically generated cationic species are capable to trigger additional reactions other than oxidative polymerization such as substitution reactions. Strong electrophilic character of generated carbocations in the presence of nucleophilic groups that are intact with the catalyst systems can create novel pathways to synthesize complex polymeric structures [150–152]. Conventional S_N1 type reactions rely on strong Lewis acids, solvents with high polarity or high temperatures to yield electrophiles to realize reactions. Recent approaches involve successive generation of radical species, oxidation to the respective carbocations and the attack of the nucleophile to establish the polymerization [153–158], thus employing a safer work environment, greener and energy-efficient reaction conditions [159,160]. Our research group has expanded the approach to dimanganese decacarbonyl chemistry by benefiting its halogen affinity. $Mn_2(CO)_{10}$ is known for its ability to abstract halogens homolytically from organo-



Scheme 15. Photochemically and thermally induced FRPCP via (a) benzophenone, (b) DMPA and (c) PAT.



Scheme 16. Visible light induced FRPCP (a) acyl germanes and (b) phosphine oxides, (c) acylsilanes (d) dimanganese decacarbonyl.



Scheme 17. Light induced cationic FP of epoxy monomers using diphenyliodonium salt as oxidizing agent and benzopinacol as thermal and photochemical radical generator.

halides to yield organo radicals [161–165]. Hence, visible light can be directly implemented in the radical generation step. Benzylic bromines were used in order to obtain faster radical generation steps. DPI was used in order to oxidize the generated radicals to their respective carbocations. Generated bifunctional electrophiles are capable to be attacked by nucleophiles. Polyol derivatives were used as nucleophiles to obtain linear or crosslinked polyethers by step-growth mechanism (Scheme 19). The effect of using chlorine homologues or initiators with steric hindrance was studied by our research group collaborating with Ortyl group in 3D-printing experiments [166]. Slower initiation step negatively affects the dispersity (*Đ*), molecular weights and also the printout resolution [167–169].

Furthermore, decreasing the strength of the nucleophile to synthesize complex macromocular structures was examined. Photochemical Friedel-Crafts type electrophilic aromatic substitution reactions were successfully applied to synthetic polymer chemistry by our research group [151,152]. Conventional aromatic substitutions mainly involve high temperatures or strong Lewis acids as catalysts to enable weak aromatic nucleophiles to attack generated carbocations [170–172]. Initial photochemical approach to achieve Friedel-Crafts reactions were reported by Naruto et al [173,174]. Main drawbacks of the approach were the small number of reactant options and requirement of UVC light. Some advanced methods were developed by several research groups to overcome the mentioned disadvantages [175–178]. Recently, we presented visible light-induced alternatives on Friedel-Crafts reactions to provide spatiotemporal control and safer reaction conditions. Typical example of this approach for the synthesis of poly(phenylene methylene) (PPM) derivatives is presented in Scheme 20.

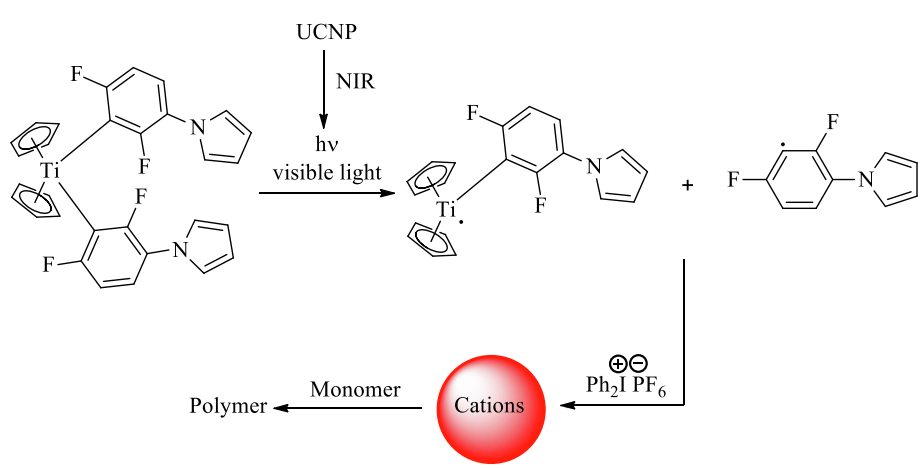
Even though the obtained polymers contain no conjugation, they exhibited fluorescence properties arising from homoconjugation at the polymer backbone [154,155]. Caseri and colleagues reported several synthetic methods to obtain PPM and investigated the physical properties of the polymer thoroughly [179–181]. In light of the observed features, we attempted to produce fluorescent block copolymers. Synthesized telechelic PPM polymers were used as macro initiators to form blocks either by FRP using methyl methacrylate (MMA) or FRPCP using cyclohexene oxide (CHO) (Scheme 21). Thus obtained block polymers were used to produce fibers by electrospinning technique.

Further approaches involve decreasing strength of the electrophile. Dihalidomethanes were deliberately selected as radical precursor since the formed corresponding radicals exhibit relatively lower stability than benzylic halide. Following initial Friedel-Crafts electrophilic aromatic substitution, the adduct benzyl halide has an increased overall reactivity. This feature allows to convert the step-growth mechanism into the chain-growth condensation polymerization mechanism. This way, the reactivities of the oligomers and polymers are increased compared to monomers (Scheme 22).

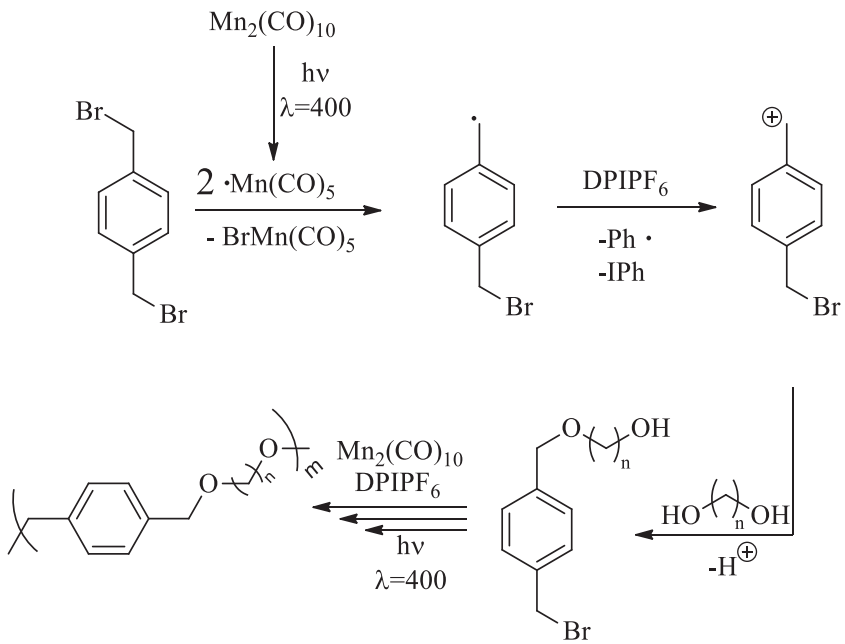
The initial studies on the chain-growth condensation mechanism were reported by Yokozawa et al [180,181]. Although the process was coined as condensative chain polymerization, it was changed to chain-growth condensation polymerization [182]. The reported visible light induced systems may lead to develop further approaches for step-growth polymerization of other monomers having hetero-aromatic rings.

5. Conclusions and Future prospects

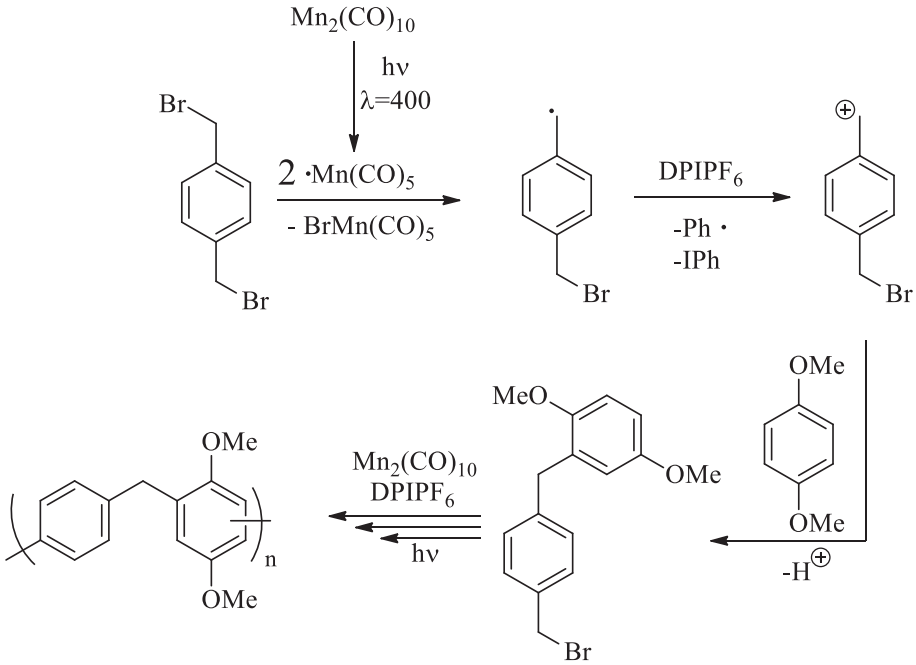
Photoinduced free-radical and cationic polymerization reactions, performed at room temperature with very little or no organic solvent and with no transition metal catalyst, are in accordance with the concept of sustainable chemistry. Many industrially important monomers such as cyclic, vinyl ethers and thiophene, thienothiophene, carbazole, pyrrole derivatives can only be photochemically polymerized in the presence of ionic species that can initiate cationic and step-growth polymerizations, respectively. The initiating methods can be classified into two main categories; direct and indirect initiating systems. In direct systems, onium salts and their structurally modified derivatives (chromophore attached) are very often used as one-component cationic photoinitiating systems under UV, near-UV and visible light polymerization. In indirect systems, photosensitizers such as typical PAHs



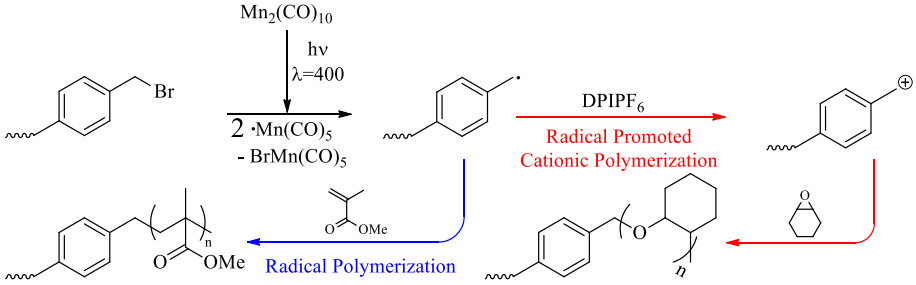
Scheme 18. NIR light induced FRPCP involving UCNP,titanocenes and DPI salt.



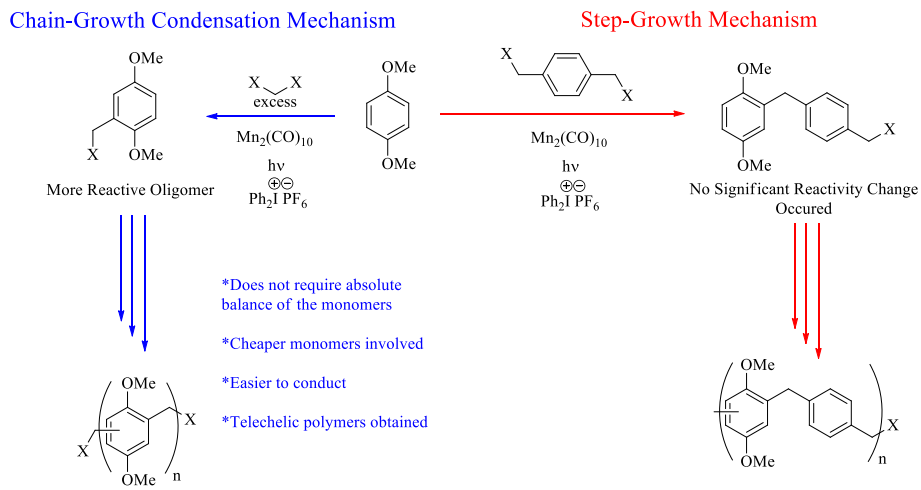
Scheme 19. Visible light induced synthesis of polyether derivatives by step-growth polymerization using FRPCP approach.



Scheme 20. Visible light induced synthesis of PPM derivatives using FRPCP approach.



Scheme 21. Synthesis of PPM-*b*-PMMA and PPM-*b*-PCHO using photo induced FRP and FRPCP techniques.



Scheme 22. Visible light induced step-growth and chain-growth condensation methods to obtain poly(phenylene methylene) derivatives.

(perylene anthracene, fluorescein…), and cyanine dyes are utilized in conjunction with onium salts to shift the excitation wavelength to visible and NIR light, respectively. Photoinduced electron transfers between these photosensitizers and onium salts can effectively initiate both the cationic polymerization of typical cationic monomers or the step-growth polymerization of a variety of industrially important heterocyclic monomers. FRPCP is another versatile method of indirect polymerization where photochemically generated radical species can be oxidized using strong oxidizing agents in order to initiate cationic polymerization. Especially, the very recent photoinduced step-growth polymerization methods developed by the group can open new pathways for synthesis of many other industrially important conjugated polymers. However, these methods failed to polymerize electron acceptor monomers. Future research will focus on the further adaption of the initiation systems and also expand the monomer scope.

Even though there is a wide range of cationic photoinitiating systems that can be utilized under a broad electromagnetic spectrum ranging from UV to NIR regions, there are constant problems regarding the purity of the final polymers obtained arising from the presence of trace amounts of photocatalysts, especially when highly colored dye photosensitizers such as cyanines are used. Particularly in the bioapplication of polymers where purity is a major concern, post-purification processes might be both time and money consuming. In order to overcome this problem, Boyer's group demonstrated the possibility of using a visible-light degradable photocatalyst, namely chlorophyll [183].

NIR light is safer than UV light, requires less energy and can penetrate through human skin making it attractive for different bioapplications. However, as explained in the introduction part, NIR light-mediated cationic polymerizations can only be reached using either relatively toxic dye photosensitizers or upconverting nanoparticles (UCNP) and there is no example of direct cationic polymerization under NIR light. This problem might be solved by the design and synthesis of new photoinitiators that can solely absorb in the NIR region. Such photoinitiators will certainly have key roles in different bioapplications including photodynamic therapy.

Modern 3D printing technologies take advantage of the temporal and spatial control ability of light-mediated polymerization reactions. Nevertheless, current 3D printing materials are mostly produced by either radical or cationic polymerization limiting the application of final polymer. To address this drawback, new hybrid photoinitiating systems can be designed and utilized for the synthesis of more complex materials that can have superior physical, chemical and mechanical properties. Especially, the very recent photoinduced step-growth polymerization methods developed by our group can open new pathways for synthesis of many other industrially important conjugated polymers using 3D printing technologies. Future research will focus on the adaption of the developed photoinitiating systems for the fabrication of novel 3D printed materials.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This paper is dedicated to the memory of our dear professor Yusuf Yagci, who passed away while this paper was being peer-reviewed.

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